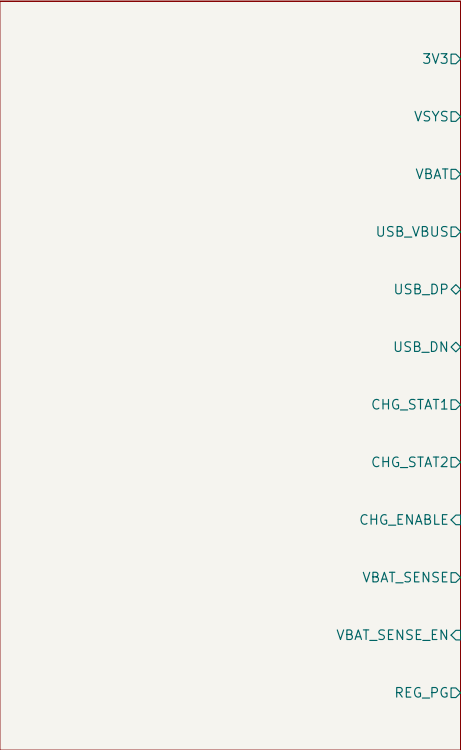


Cicala Rev B FIT STUDY system block diagram

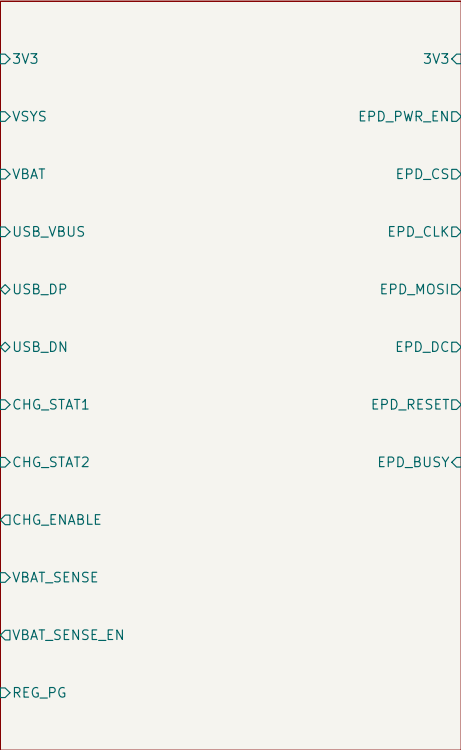
Power flow: USB-C 5 V or the protected cell -> BQ25185 power path (VSYS)
-> TPS63802 buck-boost (3V3) -> TPS22917 load switches, one for the battery divider and one for the panel and its boost network.

Power and USB



File: sheets/battery_charger.kicad_sch

Controller and user I/O



File: sheets/esp32_core.kicad_sch

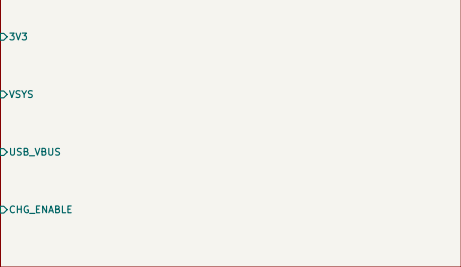
E-paper display



File: sheets/epaper.kicad_sch

GND is a global net and is not drawn here.

Charge temperature guard



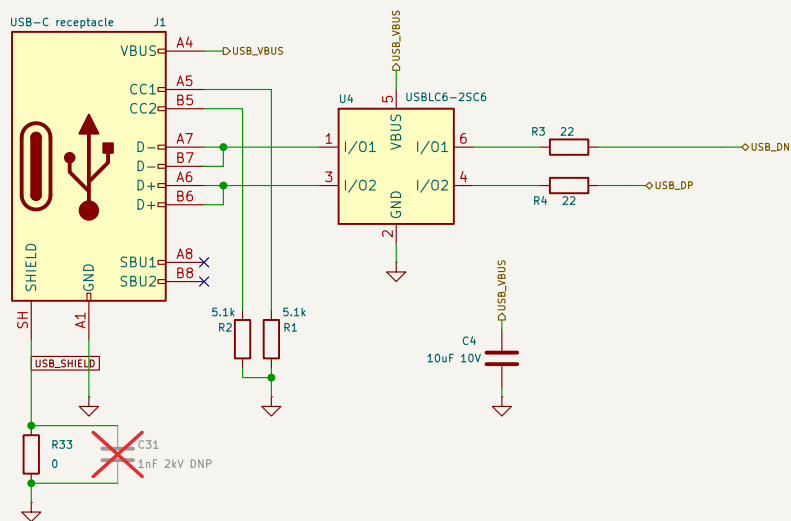
File: sheets/charge_temperature.kicad_sch

UNROUTED FIT STUDY – no firmware target / no fabrication release

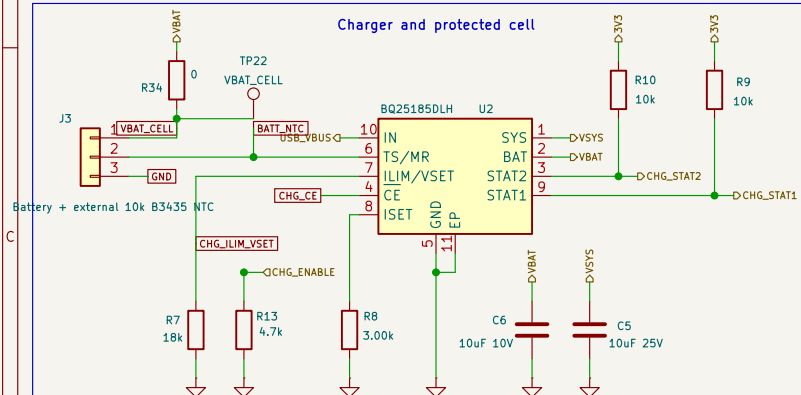
Cicala		
Sheet: /		
File: cicala_rev_b.kicad_sch		
Title: Cicala Rev B FIT STUDY schematic overview		
Size: A4	Date: 2026-09-07	Rev: Rev B FIT – NOT FOR FA
KiCad E.D.A. 10.0.5		Id: 1/5

USB-C provides 5 V and native USB.
BQ25185 charges the protected cell; TPS63802 regulates 3.3 V.

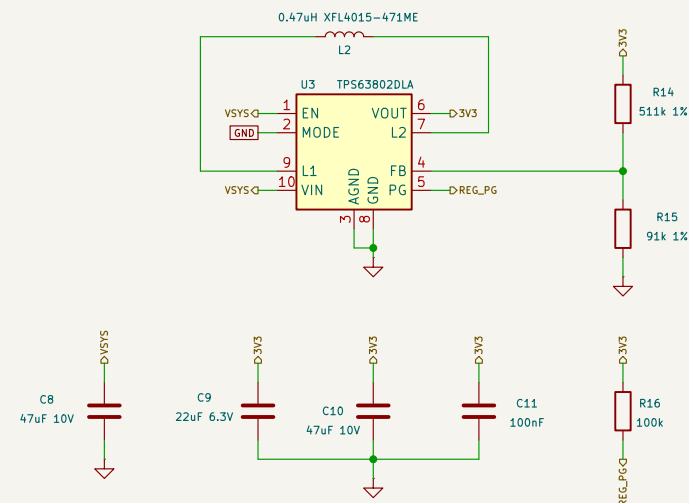
USB-C and protection



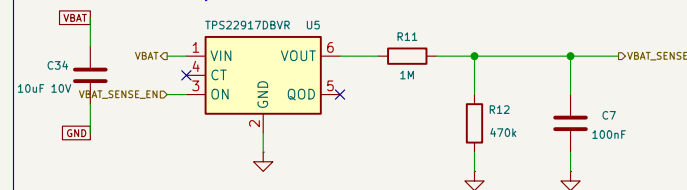
Charger and protected cell



3.3 V buck–boost



Switched battery measurement



Cicala

Sheet: /Power and USB/
File: battery_charger.kicad_sch

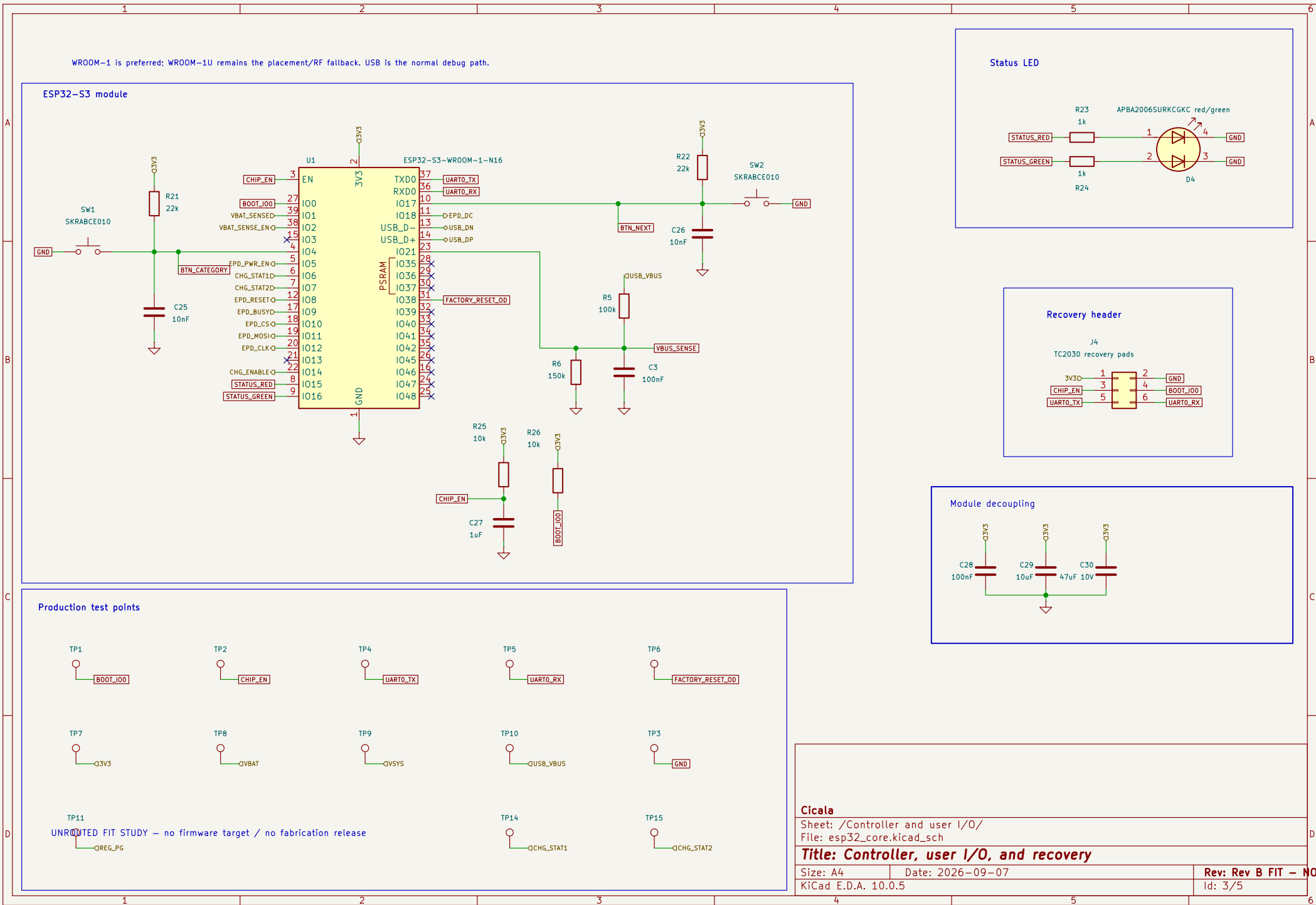
Title: *Power, charging, and USB*

Size: A4	Date: 2026-09-07
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KiCad E.D.A. 10.0.5

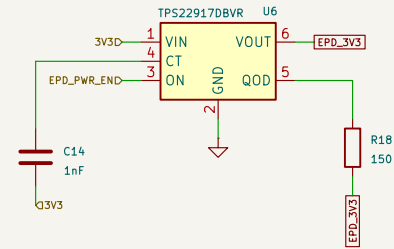
Rev: Rev B FIT – NOT FOR FA

Id: 2/5

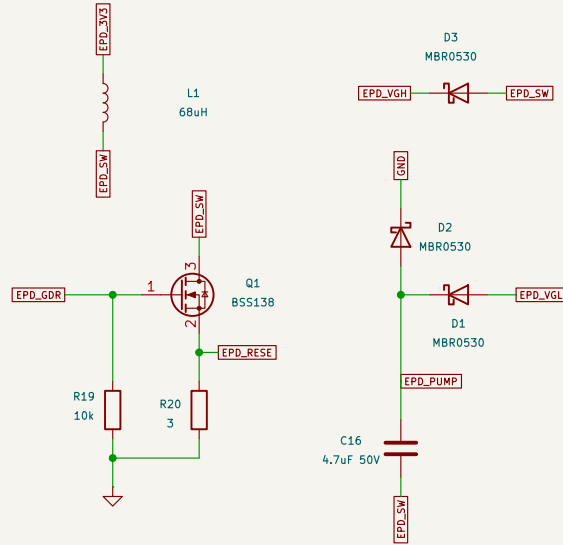


The load switch removes panel and boost power between refreshes.
Boost network follows Good Display's reference.

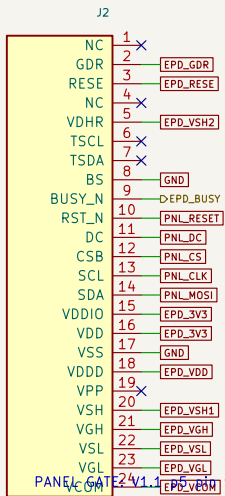
Switched panel supply



V1.1 pin table: pins 1/4 NC, pin 5 VDHR. HAT boost: 68uH / BSS138 / 3R.



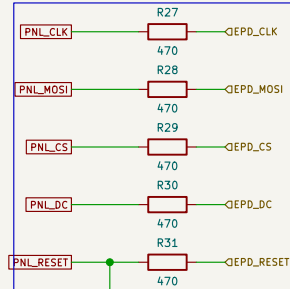
Panel FPC



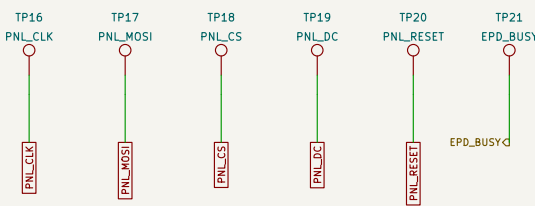
PANEL pin table conflicts with p16 circuit (pins 1/4/5). Boost values unqualified.

Waveshare 3.52in V1.1 UC8253

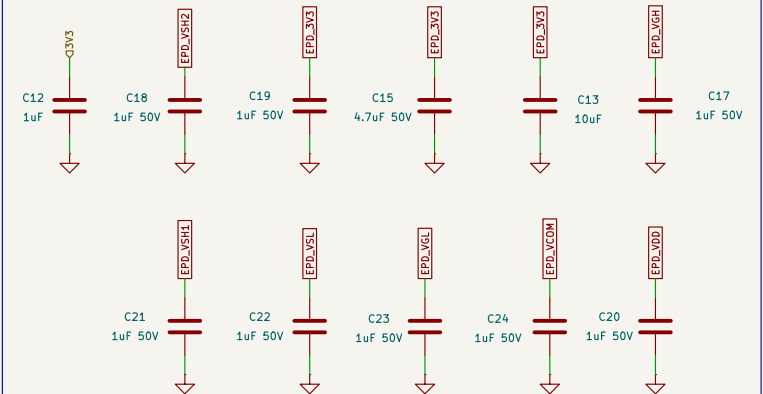
Bus isolation



Panel bus test points



Panel reservoir capacitors



Cicala

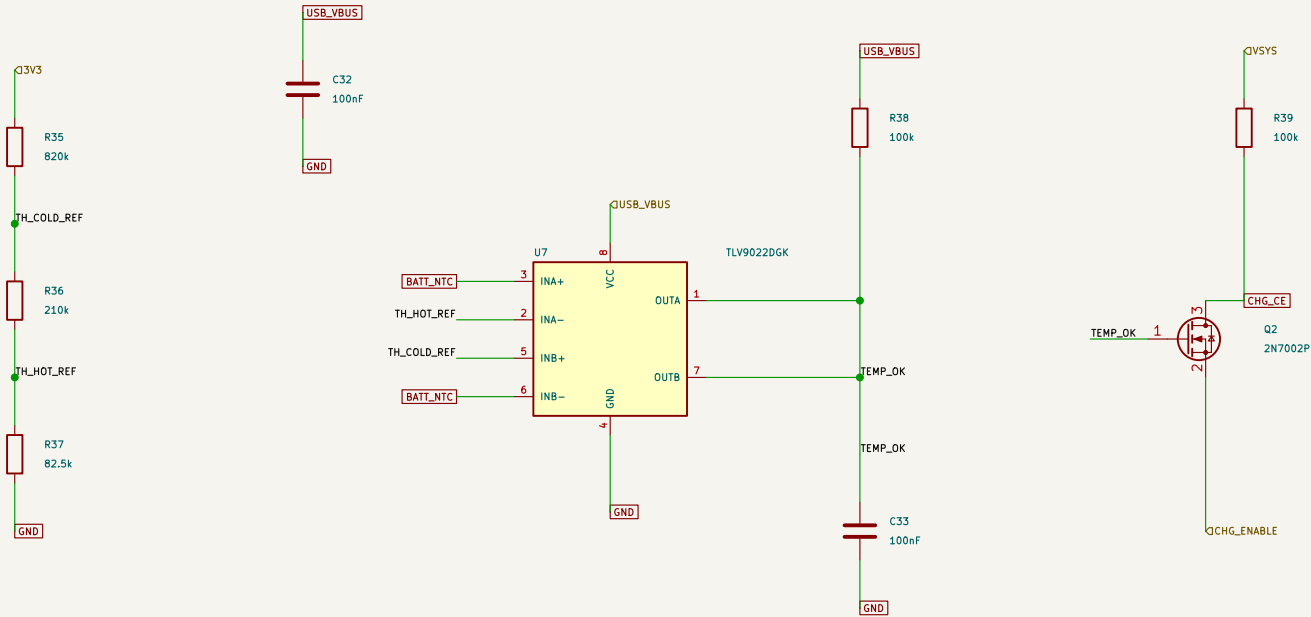
Sheet: /E-paper display/
File: epaper.kicad_sch

Title: Waveshare_352_V11_UC8253 e-paper interface and boost

Size: A4 Date: 2026-09-07
KiCad E.D.A. 10.0.5

Rev: Rev B FIT - NOT FOR FA
Id: 4/5

Independent battery temperature veto



J3.2 / BATT_NTC remains connected to BQ25185 TS/MR.
Semitec 103AT-2 is bonded to the protected cell and returns to J3.3.

Both comparator outputs must release TEMP_OK to permit charging.
An open or shorted thermistor disables charging. C33 delays permission at USB insertion.
Firmware CHG_ENABLE high disables charging; R13 supplies the low default through Q2.

3V3 sets the comparison thresholds; U7 and the gate pull-up run only from USB.
The 1.1125 Mohm reference ladder draws 2.97 uA from 3V3.

UNROUTED FIT STUDY – no firmware target / no fabrication release

Cicala		
Sheet: /Charge temperature guard/ File: charge_temperature.kicad_sch		
Title: Hardware charge temperature guard		
Size: A4	Date: 2026-09-07	Rev: Rev B FIT – NOT FOR FA
KiCad E.D.A. 10.0.5		Id: 5/5